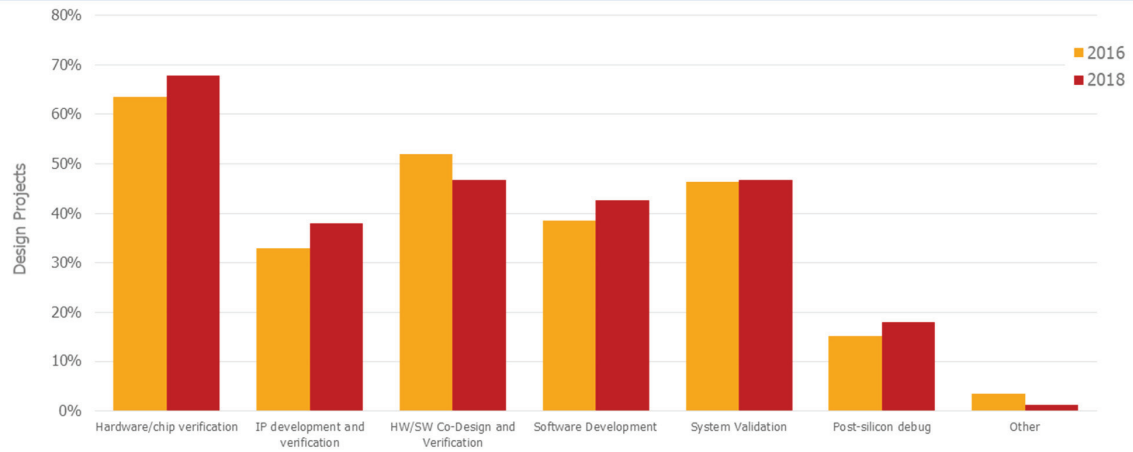


Design Verification

FPGA-based ASIC-Prototyping

Why was FPGA prototyping used on the project



Why was FPGA prototyping used on the project

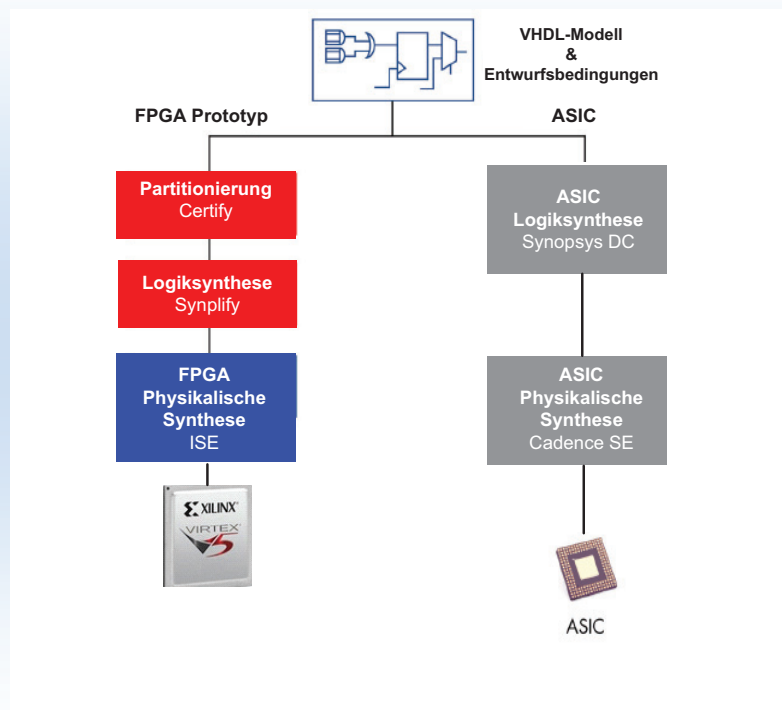
Source: Wilson Research Group and Mentor, A Siemens Business, 2018 Functional Verification Study

© Mentor Graphics Corporation

* Multiple answers possible

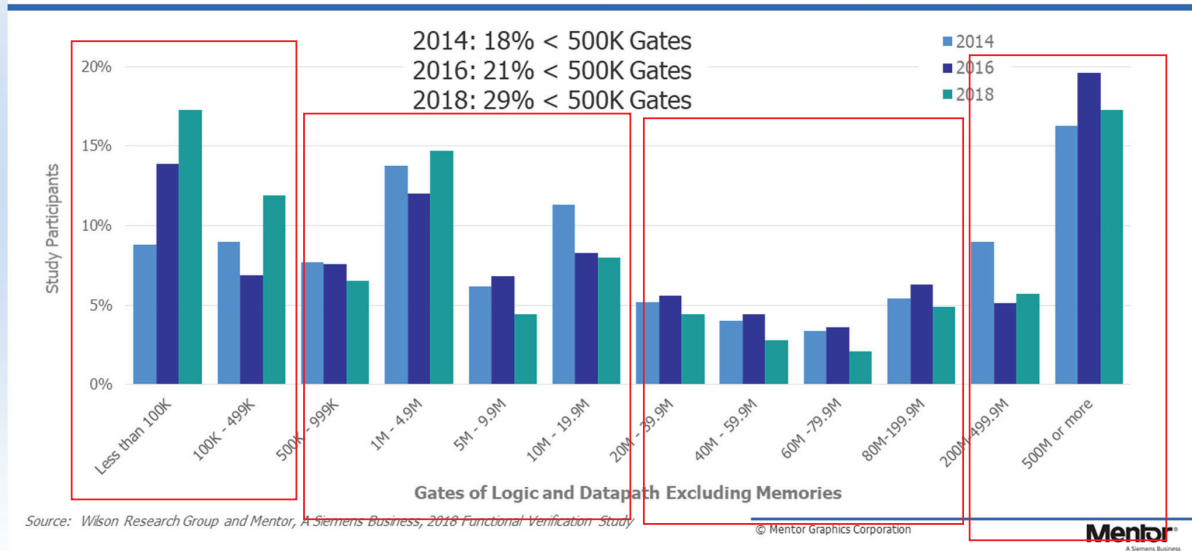
Mentor
A Siemens Business

Design-Flow: FPGA-based Prototyping



FPGA-based ASIC Prototyping

ASIC Study Participation by Gate Count (Design Size)



Prof. Schumann

VLSI Design and Testing

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FPGA-based ASIC Prototyping

Gate count FPGA

Xilinx Virtex-6 = 760K LC's = 4,560,000 two input ASIC NAND gate equivalents, (**~4.5 million**)

Xilinx Virtex-7 = 2000K LC's = 12,000,000 two input ASIC NAND gate equivalents, (**~12 million**)

Xilinx UltraScale = 4400K LC's = 26,400,000 two input ASIC NAND gate equivalents (**~26.4 million**)

Prof. Schumann

VLSI Design and Testing

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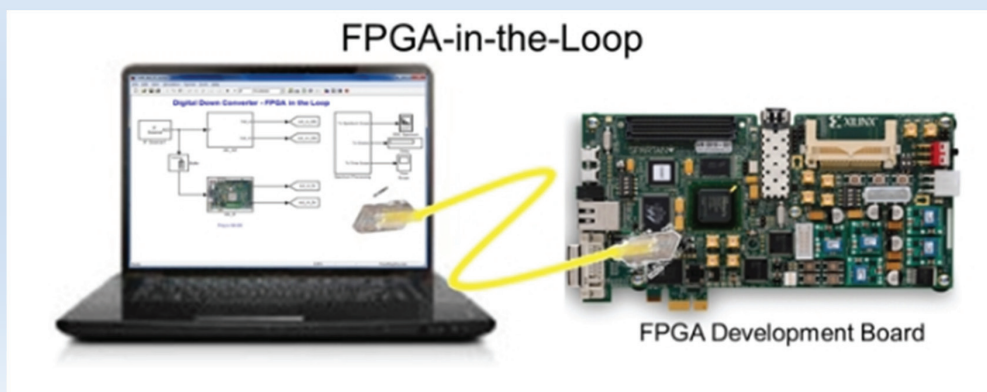
FPGA-based ASIC Prototyping

Multi-FPGA Prototyping Boards

Company	System	FPGA	No. of ASIC gates
ONIX	ONIX Platform	Xilinx Virtex UltraScale	180 M
Cadence	Protium S1	Xilinx Virtex UltraScale	200 M
Synopsys	HAPS-80	Xilinx Virtex UltraScale	1.6 B
Aldec	HES	Xilinx Virtex UltraScale	20 M-80 M
S2C	Prodigy	Xilinx/Intel	100 M

FPGA-in-the-loop Simulation

Simulink model is used to drive FPGA input stimuli and to analyze the output of the FPGA



FPGA-in-the-loop Simulation

Model-based Design to shorten HDL creation

